24. Water, rivers, lakes and sea — currents, crossings, floods and survival in water

Oracle OS guide. Based on RNLI ("Float to Live"), NOAA (rip currents), US Coast Guard (Boat Crew Seamanship, Rescue and Survival), SOBRASA (aquatic emergencies) and US Army FM 3-05.70 (crossings) — in the library **with illustrations**. For making water safe to drink see "Drinking water".

Cold water shock — "float to live"

- Falling into water below 15 °C causes **gasping and uncontrolled breathing** in the first minute: that is when many people drown.
- **Don't swim straight away. Lean back, tilt your head back, spread arms and legs, and float for 60–90 seconds** until your breathing is under control. Then swim to safety or call for help.
- With a life jacket, waiting for rescue: **HELP position** (knees to chest, arms crossed) or as a group **huddle in a circle** to keep heat. Keep clothes on (insulation).
- After pulling someone from cold water, treat hypothermia ("First aid" guide); keep them lying down and watched — they can deteriorate.

Rip currents at the beach

- Spot them: a **darker, calmer** channel between breaking waves, foam or debris moving out to sea, choppy brown water.
- If caught: **don't swim against it**. Float, raise an arm and shout. Swim **parallel to the shore** until out of it (usually 10–30 m wide), then angle back in with the waves.
- To help someone: don't go in. Throw something that floats, call 999/112 and ask for the Coastguard (UK).

Tides, beaches and coast

- Check tide tables; on wide sands and estuaries the tide can come in faster than you can walk. Always know each beach's exit.
- Mud and quicksand: don't struggle; lean back, spread your weight, move slowly backwards. Call 999/112.
- Bigger waves come without warning: keep back from rock and pier edges, never turn your back on the sea.

Rivers and moving water

- **15 cm of fast water can knock an adult over; 30–60 cm can float a car.** In floods: **turn around, don't drown**. Never drive through flooded roads.
- If you fall into a fast river: **defensive swimming position** — on your back, feet first and at the surface (pointing downstream), use your feet to fend off rocks, arms to steer. **Don't try to stand** in fast water above the knees: your foot can wedge between rocks and the current pushes you under.
- **Strainers** (fallen trees, grilles, debris on piers): water passes through, people don't. If heading into one, turn onto your front and swim hard **over** it.

- **Weirs and low dams**: the recirculating "hydraulic" below traps people and boats. Never approach from below or above. To escape: dive deep and exit underneath, with the current.

Crossing a river (only if there is no alternative)

1. Look for a bridge, known ford, or the **widest, braided** section (shallower and slower). Avoid outside bends and just above rapids, drops or obstacles.
2. Check depth and force: if it is over the knee and fast, **don't cross**.
3. Unclip pack hip belt and chest strap (so you can ditch it), keep shoes on.
4. Use a **strong pole as a third leg** on the upstream side; move sideways facing upstream, short steps, one point of contact moving at a time.
5. In a group: **in line with the current** (strongest upstream, others holding the waist of the one in front) or **huddle/triangle** with arms linked.
6. Never tie someone to a rope for a crossing: a person on a tight rope in current gets pulled under.

Lakes and ice

- Safe ice to walk on: **at least 10 cm of clear, hard ice**. White, grey, snow-covered ice or ice near stream inlets is weaker.
- If you fall through: turn toward where you came from (that ice held you), arms on the ice, **kick like swimming** and pull yourself out horizontally; **roll** away from the hole. Don't stand until on safe ice.

Survival at sea

- **Stay with the boat** (even capsized): bigger and easier to spot. Wear a life jacket, warm and waterproof clothing.
- **Never drink seawater or urine.** Ration fresh water (without water, don't eat — digestion uses water). Catch rain on tarps; solar still in a bucket.
- Shelter from sun and wind; stay dry; seasickness dehydrates — lie down and look at the horizon.
- Signals: VHF channel 16 ("MAYDAY" ×3, vessel name, position, problem, people aboard), flares, mirror, whistle, bright clothing. EPIRB/PLB if you have one.
- Only abandon ship to a liferaft **when the boat is really sinking** ("step up into the raft", don't jump into the water).

Drowning — recognise and rescue

- Drowning people **rarely shout or wave**: upright, head back, mouth at water level, not moving forward. Ask "Are you OK?" — no answer, act.
- **Reach, throw, row, go**: reach with a pole/rope/towel while lying down; throw something that floats; use a boat; only go in trained and with buoyancy.
- Out of the water: unresponsive and not breathing normally → **5 rescue breaths then CPR 30:2** (see "First aid"). Even if they recover, they must see a doctor.

Small boats and rafts

- Weight low and central, three points of contact when boarding, always a life jacket. Don't overload.
- **Emergency raft**: sealed empty drums/jerrycans or dry logs, fixed to a pole frame with square lashings ("Ropes and knots"). Test in calm shallow water.

Lakes and rivers as a resource

- Surface water must be **treated** before drinking (boil 1 minute). Collect upstream of settlements, animals and latrines; never drink water with odd foam, chemical smell or blue-green algae (cyanobacteria — boiling doesn't destroy their toxins).
- Wash yourself and clothes **60 m from the water**, using a container, so the source stays clean.